Face Effects

Note: There is relevant running code on Raspberry Pi and Jetson Nano, but due to differences in motherboard performance, running it may not be as smooth. The accompanying virtual machine also has the running environment and programs installed. If the experience on the motherboard is not satisfactory, you can remove the camera, insert it into the virtual machine, and connect the camera device in the virtual machine to run the corresponding program.

6.1 Introduction

MediaPipe is a data stream processing machine learning application development framework developed and open-sourced by Google. It is a graph-based data processing pipeline used to build applications that use multiple forms of data sources, such as video, audio, sensor data, and any time series data. MediaPipe is cross-platform and can run on embedded platforms (Raspberry Pi and similar), mobile devices (iOS and Android), workstations and servers, with support for mobile GPU acceleration. MediaPipe provides cross-platform, customizable ML solutions for real-time and streaming media. The core framework of MediaPipe is implemented in C++ and provides support for Java and Objective C. The main concepts of MediaPipe include Packet, Stream, Calculator, Graph, and Subgraph.

MediaPipe features:

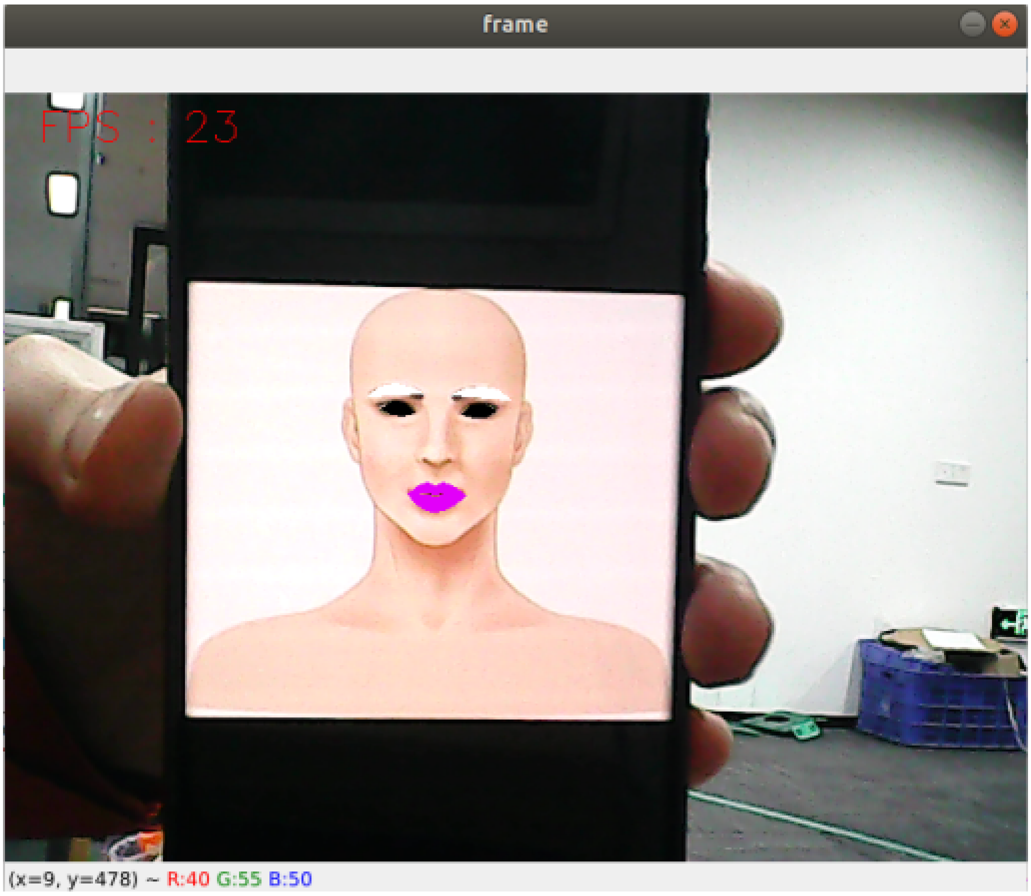
- End-to-end acceleration: Built-in fast ML inference and processing accelerate even on ordinary hardware.
- Build once, deploy anywhere: Unified solutions work on Android, iOS, desktop/cloud, web, and IoT.
- Ready-to-use solutions: Cutting-edge ML solutions that showcase all framework features.
- Free and open source: Framework and solutions under Apache 2.0, fully extensible and customizable.

6.2 Face Effects

6.2.1 Launch

Enter in terminal:

```
ros2 run dofbot_mediapipe 06_FaceLandmarks
```



6.2.2 Source Code

Source location: /home/yahboom/dofbot_ws/src/dofbot_mediapipe/dofbot_mediapipe/06_FaceLandmarks.py

```
#!/usr/bin/env python3
# encoding: utf-8

import time
import dlib
import os
import cv2 as cv
import numpy as np
import rclpy
from rclpy.node import Node
from sensor_msgs.msg import Image
from cv_bridge import CvBridge

class FaceLandmarks(Node):
    def __init__(self, dat_file):
        super().__init__('face_landmarks')
        self.publisher_ = self.create_publisher(Image,
'face_landmarks_detected', 10)
        self.timer = self.create_timer(0.1, self.timer_callback)
        self.bridge = CvBridge()
        self.hog_face_detector = dlib.get_frontal_face_detector()
        self.dlib_facelandmark = dlib.shape_predictor(dat_file)
        self.capture = cv.VideoCapture(0, cv.CAP_V4L2)
        self.capture.set(cv.CAP_PROP_FRAME_WIDTH, 640)
        self.capture.set(cv.CAP_PROP_FRAME_HEIGHT, 480)
        self.pTime = 0
```

```python
    def timer_callback(self):
        ret, frame = self.capture.read()
        if not ret:
            self.get_logger().error('Failed to capture frame')
            return
        frame = self.get_face(frame, draw=False)
        frame = self.prettify_face(frame, eye=True, lips=True, eyebrow=True, draw=True)

        # Calculate FPS
        cTime = time.time()
        fps = 1 / (cTime - self.pTime)
        self.pTime = cTime

        # Display FPS on frame
        cv.putText(frame, f'FPS: {int(fps)}', (20, 30), cv.FONT_HERSHEY_SIMPLEX, 0.9, (0, 0, 255), 1)

        msg = self.bridge.cv2_to_imgmsg(frame, "bgr8")
        self.publisher_.publish(msg)

        # Show the frame with face detection
        cv.imshow('Face Landmarks', frame)
        if cv.waitKey(1) & 0xFF == ord('q'):
            self.capture.release()
            cv.destroyAllWindows()
            rclpy.shutdown()

    def get_face(self, frame, draw=True):
        gray = cv.cvtColor(frame, cv.COLOR_BGR2GRAY)
        self.faces = self.hog_face_detector(gray)
        for face in self.faces:
            self.face_landmarks = self.dlib_facelandmark(gray, face)
            if draw:
                for n in range(68):
                    x = self.face_landmarks.part(n).x
                    y = self.face_landmarks.part(n).y
                    cv.circle(frame, (x, y), 2, (0, 255, 255), 2)
                    cv.putText(frame, str(n), (x, y), cv.FONT_HERSHEY_SIMPLEX, 0.6, (0, 255, 255), 2)
        return frame

    def get_lmList(self, frame, p1, p2, draw=True):
        lmList = []
        if len(self.faces) != 0:
            for n in range(p1, p2):
                x = self.face_landmarks.part(n).x
                y = self.face_landmarks.part(n).y
                lmList.append([x, y])
                if draw:
                    next_point = n + 1
                    if n == p2 - 1: next_point = p1
                    x2 = self.face_landmarks.part(next_point).x
                    y2 = self.face_landmarks.part(next_point).y
                    cv.line(frame, (x, y), (x2, y2), (0, 255, 0), 1)
        return lmList

    def get_lipList(self, frame, lipIndexlist, draw=True):
```

```
        lmList = []
        if len(self.faces) != 0:
            for n in range(len(lipIndexlist)):
                x = self.face_landmarks.part(lipIndexlist[n]).x
                y = self.face_landmarks.part(lipIndexlist[n]).y
                lmList.append([x, y])
                if draw:
                    next_point = n + 1
                    if n == len(lipIndexlist) - 1: next_point = 0
                    x2 = self.face_landmarks.part(lipIndexlist[next_point]).x
                    y2 = self.face_landmarks.part(lipIndexlist[next_point]).y
                    cv.line(frame, (x, y), (x2, y2), (0, 255, 0), 1)
        return lmList

    def prettify_face(self, frame, eye=True, lips=True, eyebrow=True,
draw=True):
        if eye:
            leftEye = self.get_lmList(frame, 36, 42)
            rightEye = self.get_lmList(frame, 42, 48)
            if draw:
                if len(leftEye) != 0: frame = cv.fillConvexPoly(frame,
np.array(leftEye, dtype=np.int32), (0, 0, 0))
                if len(rightEye) != 0: frame = cv.fillConvexPoly(frame,
np.array(rightEye, dtype=np.int32), (0, 0, 0))
        if lips:
            lipIndexlistA = [51, 52, 53, 54, 64, 63, 62]
            lipIndexlistB = [48, 49, 50, 51, 62, 61, 60]
            lipsUpA = self.get_lipList(frame, lipIndexlistA, draw=True)
            lipsUpB = self.get_lipList(frame, lipIndexlistB, draw=True)
            lipIndexlistA = [57, 58, 59, 48, 67, 66]
            lipIndexlistB = [54, 55, 56, 57, 66, 65, 64]
            lipsDownA = self.get_lipList(frame, lipIndexlistA, draw=True)
            lipsDownB = self.get_lipList(frame, lipIndexlistB, draw=True)
            if draw:
                if len(lipsUpA) != 0: frame = cv.fillConvexPoly(frame,
np.array(lipsUpA, dtype=np.int32), (249, 0, 226))
                if len(lipsUpB) != 0: frame = cv.fillConvexPoly(frame,
np.array(lipsUpB, dtype=np.int32), (249, 0, 226))
                if len(lipsDownA) != 0: frame = cv.fillConvexPoly(frame,
np.array(lipsDownA, dtype=np.int32), (249, 0, 226))
                if len(lipsDownB) != 0: frame = cv.fillConvexPoly(frame,
np.array(lipsDownB, dtype=np.int32), (249, 0, 226))
        if eyebrow:
            lefteyebrow = self.get_lmList(frame, 17, 22)
            righteyebrow = self.get_lmList(frame, 22, 27)
            if draw:
                if len(lefteyebrow) != 0: frame = cv.fillConvexPoly(frame,
np.array(lefteyebrow, dtype=np.int32), (255, 255, 255))
                if len(righteyebrow) != 0: frame = cv.fillConvexPoly(frame,
np.array(righteyebrow, dtype=np.int32), (255, 255, 255))
        return frame

def main(args=None):
    rclpy.init(args=args)
    current_dir_path = os.path.dirname(__file__)
    dat_file = os.path.join(current_dir_path,
"/home/yahboom/dofbot_ws/src/dofbot_mediapipe/dofbot_mediapipe/file/shape_predic
tor_68_face_landmarks.dat")
```

```python
    face_landmarks = FaceLandmarks(dat_file)
    rclpy.spin(face_landmarks)
    face_landmarks.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```